Kasra Fouladi

Research Interests: Imitation Learning, POMDPs, Representation Learning, Continual Learning, Interpretable Machine Learning

B.Sc. Student in Computer Science
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Education

B.Sc. in Computer Science **2023 – 2027 (Expected)**
Iran University of Science and Technology (IUST) *Tehran, Iran*
GPA: 93.15/100
Relevant Coursework (grades 95+):

- **Mathematics & Theory:** Mathematical Analysis (100/100), Set Theory (100/100), Mathematical Foundations (97.5/100), Linear Optimization (97.5/100), Probability (96.5/100), Combinatorics (95/100), Fundamentals of Computation Theory (95/100), Ordinary Differential Equations (95/100).
- **CS & Research:** Data Structures (100/100), Design and Analysis of Algorithms (100/100), Research Methodology and Report Writing (100/100; included practical work), Internship (100/100; research internship at IPM).

Research Experience

Research Intern **Jun 2026 – Sep 2026**
Institute for Research in Fundamental Sciences (IPM) *Tehran, Iran*

- Investigating decoupled representation learning and theoretical credit assignment in partially observable environments (POMDPs).
- Investigating continual learning and behavioral repair in partially observable environments, with a focus on catastrophic forgetting and targeted post-hoc correction.

Research Publications (Preprints)

Interpretability-Guided Bi-objective Optimization: Aligning Accuracy and Explainability *Jan 2026*
K. Fouladi (corresponding), H. Rahmani. arXiv: arXiv:2601.00655
Proposed a bi-objective training framework that jointly optimizes predictive accuracy and model explainability, using interpretability signals to guide the optimization toward solutions that are both accurate and human-understandable.

StrikeForce: A 2D Partially Observable Battle-Arena Testbed for Imitation Learning and Multi-Agent Research *Jul 2026*
Sole author. DOI: doi.org/10.13140/RG.2.2.24446.96326 | GitHub: github.com/bistoyek21-ric/StrikeForce
Built an efficient open-source research environment/testbed for RL/IL, with deterministic replay, networked multiplayer, and a pluggable agent interface.

No Back-Propagation Through Time via Future Action Prediction *Jul 2026*
Sole author. DOI: doi.org/10.13140/RG.2.2.18971.43049 | GitHub: github.com/kasrafouladi/StrikeForce-No-BPTT-via-FAP
Proposed a decoupled imitation-learning architecture that removes BPTT from the perception backbone: a Future-Action Predictor backbone trained only on a multi-step future-action loss, feeding a stop-gradient boundary into a bounded-window Transformer aggregator. Proved the backbone's credit-assignment path has length 1 (no temporal Jacobian product), derived a Hedge no-regret guarantee for the aggregation step, and gave a general bias–variance analysis of the aggregation. Validated on the StrikeForce environment against a no-aggregator and a linear-fusion baseline.

Targeted Behavioral Repair with Contaminated Anchors *Sep 2026*
Sole author. DOI: doi.org/10.13140/RG.2.2.11421.68328 | GitHub: github.com/kasrafouladi/StrikeForce-No-BPTT-via-FAP
Studied post-hoc correction of an identified behavioral failure in a converged policy via a combined EWC (parameter-space) + KD (output-space) objective, formalizing the *contaminated-anchor* setting where the anchor set may itself contain the failure. Derived a local multi-step convergence result showing the repair converges to a curvature-weighted equilibrium displacement, and characterized how anchor contamination perturbs that equilibrium as $O(\rho)$. Applied to repair a behavioral-cloning policy trained on StrikeForce.

Work in Progress

TINAS: Transformer Interaction among N Agents with Shared Parameters *(B.Sc. Final thesis) Oct 2026*
Sole author. DOI: doi.org/10.13140/RG.2.2.24504.71689 | GitHub: github.com/kasrafouladi/StrikeForce-No-BPTT-via-FAP
Proposes a communication-learning framework for MARL that decouples credit-assignment horizon from graph-memory depth. Messages are trained by *future-gradient injection*: the sender's loss at time t is built from the receiver's loss at $t+1$ via the sender's positional encoding, avoiding backpropagation through time. Proves a gradient-equivalence lemma, bounds the straight-through-estimator bias, and establishes an $O(1/\sqrt{K})$ convergence rate. Achieves $\Omega(L)$ per-step memory reduction versus DIAL/BPTT using a two-pass backward algorithm at $O(T)$ cost. Implementation on a frozen behavioral-cloning backbone for the StrikeForce testbed; code publicly available. Technical report, not yet peer-reviewed.

Teaching Experience

Teaching Assistant (Iran University of Science and Technology):
Basic Programming – Fall 2024; Combinatorics – Spring 2025; General Mathematics 1, Fundamentals of Computation Theory, Quantum Computing – Fall 2025; Theory of Computation – Spring 2026.

Technical Skills

Languages & Frameworks: Python, C/C++, LaTeX, SQL; PyTorch, TensorFlow, Scikit-learn, LibTorch, SFML, Django, Flask.
Tools & Platforms: Git, Linux, Docker, CMake, Overleaf, Shell scripting.

Languages

Persian: Native | **English:** Fluent